

Yaswanth Chittepu

PhD Student, Computer Science, University of Massachusetts Amherst · Amherst, MA

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Research Interests

My research focuses on designing **safe and efficient reinforcement learning algorithms**: alignment of large language models (LLMs) with high-confidence guarantees and risk control, reward overoptimization, offline RL, imitation learning, inverse RL, and LLM agents.

Education

University of Massachusetts Amherst

DOCTOR OF PHILOSOPHY, COMPUTER SCIENCE | GPA: 3.97/4.0

Amherst, MA, USA

Expected 2027

- Advised by **Prof. Scott Niekum** (SCALAR Lab)

University of Massachusetts Amherst

MASTER OF SCIENCE, COMPUTER SCIENCE | GPA: 3.97/4.0

Amherst, MA, USA

2021

Indian Institute of Technology (IIT), Kharagpur

BACHELOR OF TECHNOLOGY, MECHANICAL ENGINEERING

Kharagpur, India

2016

Publications

* Equal contribution.

- Convex-Concave Reinforcement Learning*
Shripad V. Deshmukh, **Yaswanth Chittepu**, Dhawal Gupta, Philip S. Thomas, Scott Niekum | Preprint 2026 [\[paper\]](#) [\[code\]](#)
- Learning Compact State Abstractions for LLM Agents*
Yaswanth Chittepu, et al. | Under review, 2026
- Safe Inference-Time Alignment via Lagrangian Reward Augmentation*
Yaswanth Chittepu, Ativ Joshi, Sohini Chintala, Scott Niekum | Conference on Language Modeling (COLM) 2026 [\[paper\]](#) [\[website\]](#)
- Safe RLHF Beyond Expectation: Stochastic Dominance for Universal Spectral Risk Control*
Yaswanth Chittepu, Ativ Joshi, Rajarshi Bhattacharjee, Scott Niekum | Preprint 2026 [\[paper\]](#)
- ML-Tool-Bench: Tool-Augmented Planning for ML Tasks*
Yaswanth Chittepu, Raghavendra Addanki, Tung Mai, Anup Rao, Branislav Kveton | Preprint 2025 [\[paper\]](#)
- Adaptive Margin RLHF via Preference over Preferences*
Yaswanth Chittepu, Prasann Singhal, Greg Durrett, Scott Niekum | Preprint 2025 [\[paper\]](#)
- Reinforcement Learning from Human Feedback with High-Confidence Safety Guarantees*
Yaswanth Chittepu*, Blossom Metevier*, Will Schwarzer, Austin Hoag, Scott Niekum, Philip S. Thomas | Reinforcement Learning Conference (RLC) 2025; Reinforcement Learning Journal, vol. 6 [\[paper\]](#) [\[website\]](#) [\[code\]](#)
- Scaling Laws for Reward Model Overoptimization in Direct Alignment Algorithms*
Rafael Rafailov*, **Yaswanth Chittepu***, Ryan Park*, Harshit Sikchi*, Joey Hejna*, W. Bradley Knox, Chelsea Finn, Scott Niekum | NeurIPS 2024; also at the Models of Human Feedback for AI Alignment (MHFAIA) workshop @ ICML 2024 and the RLBrew workshop @ RLC 2024 [\[paper\]](#)

Research Experience

Adobe Research | Research Intern

COMPACT STATE ABSTRACTIONS FOR LLM AGENTS

San Jose, CA, USA

May 2026 – Aug 2026

- Formulated agent state for black-box LLM agents as a **compact abstraction of the interaction history** that preserves a fixed reference agent's task performance, connecting LLM agents to classical state abstraction.
- Represented state abstractions as **deterministic recursive programs** and discovered them with an **LLM-guided genetic algorithm**; the learned programs are **directly inspectable** and need no extra model calls at deployment.
- On ALFWorld and ScienceWorld, reduced mean policy input by **up to 71% and 57%** while preserving near-reference task success, exceeding full-history success on **4 of 10 task families** and staying within 8 points on the rest.

Adobe Research | Research Intern

TOOL-AUGMENTED PLANNING FOR ML AGENTS

San Jose, CA, USA

May 2025 – Aug 2025

- Built **ML-Tool-Bench**, a benchmark for tool-augmented ML agents with **61 specialized tools** and **15 Kaggle tasks**.
- Designed planning methods using **shaped deterministic rewards** with structured textual feedback, and **task decomposition** into sub-tasks, improving over ReAct baselines by **16.5 leaderboard percentile points**.
- Introduced **in-memory object management** (a shared scratchpad of named artifacts) to improve agent planning and workflow reliability.

Robotics Research Center, IIIT Hyderabad | Research Intern

REINFORCEMENT LEARNING FOR HUMANOID MANIPULATION

Hyderabad, India

Jul 2018 – Jul 2019

- Learned reaching and **dual-arm manipulation policies** for a humanoid with **DDPG, modified for shared action spaces** to capture coupling between motors; reached goals among obstacles with an **85% success rate**, with cubic-spline interpolation for smooth, physically executable trajectories.
- Trained a humanoid to **imitate human motion from videos**: pose estimation with Human Mesh Recovery, motion reconstruction to remove jitter and infeasible poses, and PPO in MuJoCo with a pose-similarity reward.

Research Projects

Safe Inference-Time Alignment (LARA)

SCALAR Lab, UMass Amherst

COLM 2026

- Moved the Safe RLHF constraint **from training to decoding**: dualizing the constraint reduces the search over policies to a **one-dimensional convex problem** over a **single Lagrange multiplier λ^*** .
- Calibrated λ^* by binary search on a small prompt set in **about two minutes**; the resulting reward $r - \lambda^*c$ plugs into Best-of- N , ARGS, PARGS, or CD-Q with **no weight updates**, and Best-of- N approaches **finetuned baselines**.

High-Confidence Safe RLHF (HC-RLHF)

SCALAR Lab, UMass Amherst

RLC 2025

- Proposed the **first Seldonian algorithm for RLHF**: optimize helpfulness under a pessimistic upper-confidence bound on cost, then return the model only if it passes a **held-out statistical safety test**.
- Derived a **variance-aware policy gradient** for candidate selection, **proved** that an unsafe model is returned with probability **at most δ** , and validated on Qwen2-1.5B, Qwen2.5-3B, and Llama-3.2-3B with improved harmlessness and helpfulness.

Risk-Sensitive Alignment beyond Expected Cost

SCALAR Lab, UMass Amherst

PREPRINT 2026

- Replaced the expected-cost constraint in Safe RLHF with **first-order stochastic dominance** over the full cost distribution, which controls a broad family of **spectral risk measures**; made it tractable with optimal transport and Sinkhorn iterations.

Adaptive Margin RLHF via Preference over Preferences

SCALAR Lab, UMass Amherst

PREPRINT 2025

- Proposed **DPO-PoP**, which uses preference-over-preference supervision to infer **adaptive margins, outperforming fixed and ground-truth margin baselines** on UltraFeedback; analyzed the trade-off between discriminative and generative performance and designed sampling strategies to prioritize either.

Reward Overoptimization in Direct Alignment Algorithms

SCALAR Lab, UMass Amherst

NEURIPS 2024

- Showed that direct alignment algorithms (DPO, IPO, SLiC) **degrade at high KL budgets** like classical RLHF, **begin overfitting before a single epoch completes**, and, being under-constrained, often **favor out-of-distribution responses**; smaller models overfit to simple features such as length.

Model-Free RL for Diabetes Treatment

UMass Amherst

M.S. PROJECT

- Learned **insulin-dosing policies** from continuous glucose monitoring data with DDPG, using **high-confidence off-policy improvement** to switch from the behavior policy only when the new policy is better with high confidence.

Industry Experience

athenahealth | Member of Technical Staff

PAYMENTS INFRASTRUCTURE

Remote

Jul 2021 – Mar 2022

- Designed and developed the **payments infrastructure** for bill generation and customer payment processing; led the database schemas and APIs for backend–frontend integration.

athenahealth | Software Engineering Intern

MENTOR MATCHING WEB APP

Remote

Jun 2020 – Aug 2020

- Built a React web app that matches mentors to mentees under preferences and constraints with an **optimized backtracking algorithm**, REST APIs, and SQL schemas; deployed it on **AWS** via a **Jenkins CI/CD pipeline** (Docker).

dunnhumby | Data Scientist

COMPUTER VISION AND CUSTOMER ANALYTICS

Gurugram, India

Jun 2016 – Jun 2018

- Automated text extraction from scanned receipts: **conditional GANs** to detect tables, and Hough transforms to extract bounding boxes and key-value pairs.
- Segmented Tesco stores in Central Europe with **Gaussian mixture models** (EM, PCA, BIC for model selection), improving silhouette score by **over 30%** versus k-means.

Talks and Presentations

- *Safe Inference-Time Alignment via Lagrangian Reward Augmentation* | Poster, COLM 2026, San Francisco | Oct 2026
- *Reinforcement Learning from Human Feedback with High-Confidence Safety Guarantees* | Talk and poster, RLC 2025, Edmonton | Aug 2025
- *Scaling Laws for Reward Model Overoptimization in Direct Alignment Algorithms* | Poster, NeurIPS 2024, Vancouver | Dec 2024

Awards

2026 Gold Reviewer, International Conference on Machine Learning (ICML)

Service

- **Conference reviewer:** NeurIPS 2026, ICML 2026 (**Gold Reviewer**), ICLR 2026, COLM 2026.
- **Holistic reviewer:** Reinforcement Learning Conference (RLC) 2026.
- **Journal reviewer:** Transactions on Machine Learning Research (TMLR).
- **Workshop reviewer:** RLBrew @ RLC 2024, 2025; Models of Human Feedback for AI Alignment @ ICML 2024, 2025.

Skills

Languages Python, C, C++, JavaScript/TypeScript, MATLAB

Methods RLHF (PPO, GRPO, RLOO, DPO), supervised fine-tuning (SFT), distributed multi-GPU training, LLM agents and tool use

ML / RL PyTorch, JAX, Hugging Face Transformers, TRL, DeepSpeed, Accelerate, vLLM, LangGraph, Gymnasium, Stable-Baselines

Robotics ROS, MuJoCo, Gazebo